Optimization and matching exercises

1. Give a source-reachable graph with a negative edge but no negative cycle where ordinary Dijkstra finalization returns a wrong distance.
2. On A, B, C, D with AB:1,AC:4,BC:2,BD:5,CD:3,find an MST and verify at least one edge by the cut property.
3. Build a three-job/three-worker bipartite graph failing Hall. Exhibit the deficient subset and a maximum matching.
4. Add source/sink to a unit-capacity bipartite reduction and prove that an integral maximum flow's used L→R arcs form a matching.
5. For capacities sa:3,sb:2,ab:1,at:2,bt:3,find maximum flow, minimum cut, and final residual reachable set.
6. Explain what can go wrong if Kruskal adds minimum-weight edges without cycle checks and how DSU repairs it.
7. Use source A and arcs A→B:3,A→C:1,C→B:1,B→D:2,C→D:7,D→E:0,B→E:6,with isolated F. Trace every Dijkstra distance, finalized set, and predecessor. Recover the path to E and verify every arc inequality. Would adding a negative self-loop at F alter these A-distances? Distinguish a negative cycle's existence from source reachability.
8. Prove a spanning tree T is the unique MST iff every nontree edge is strictly heavier than the heaviest edge on its T-path. For cycle A–B–C–D–A with all cycle weights one and diagonals AC, BD of weight two, list every MST and identify edges appearing in all of them.
9. Add original ba of capacity two to the text's unit-capacity sa, sb, ab, at, bt network. First augment s–a–b–t by one. List the two b→a residual records and capacities. Then augment s–b–a–t by one using either original ba or reverse ab. Give both resulting original-flow vectors, values, and optimum certificates.
10. Let a, b, c each neighbor 1,2 and d only neighbor 3. Find maximum matching, Hall deficiency, and a minimum cover from alternating reachability, listing each reachable set. Add right vertex four and edge c4;find a new maximum matching and minimum cover.
11. Left i has integer demand $d_i \geq 0$ and right j capacity $b_j \geq 0$. Allowed pairs may carry multiple units. Prove all demands are feasible iff every $S \subseteq L$ has $d(S) \leq b(N(S))$,using a full source–left–right–sink cut proof. Solve d=(2,1),b=(1,2),where the first left vertex neighbors both right vertices and the second only the first. Is the same condition sufficient if each pair is limited to one unit?

Optimization and matching exercises (continued)

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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